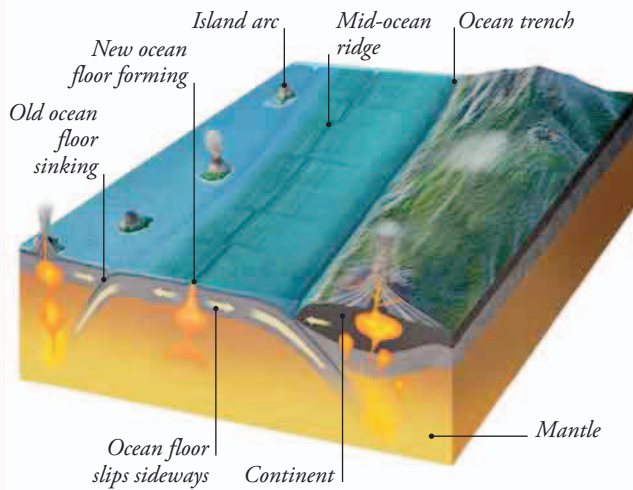


CREATION AND DESTRUCTION

New ocean floor is created in the spreading rifts of mid-ocean ridges and eventually destroyed in the subduction zones beneath ocean trenches. These changes happen at different rates in each of the world's oceans, which are constantly growing and shrinking in size.



▲ CONTINUOUS PROCESS

This illustration shows how the ocean floor forms at the dark mid-ocean ridge, moves away from the ridge, and eventually sinks back into the hot mantle beneath Earth's crust.

SEAS OF CHANGE

Over hundreds of millions of years, the Pacific Ocean has been shrinking, because its ocean floor is being destroyed in subduction zones all around the Pacific Ring of Fire. Meanwhile, the Atlantic Ocean has very few subduction zones, and has been steadily growing.

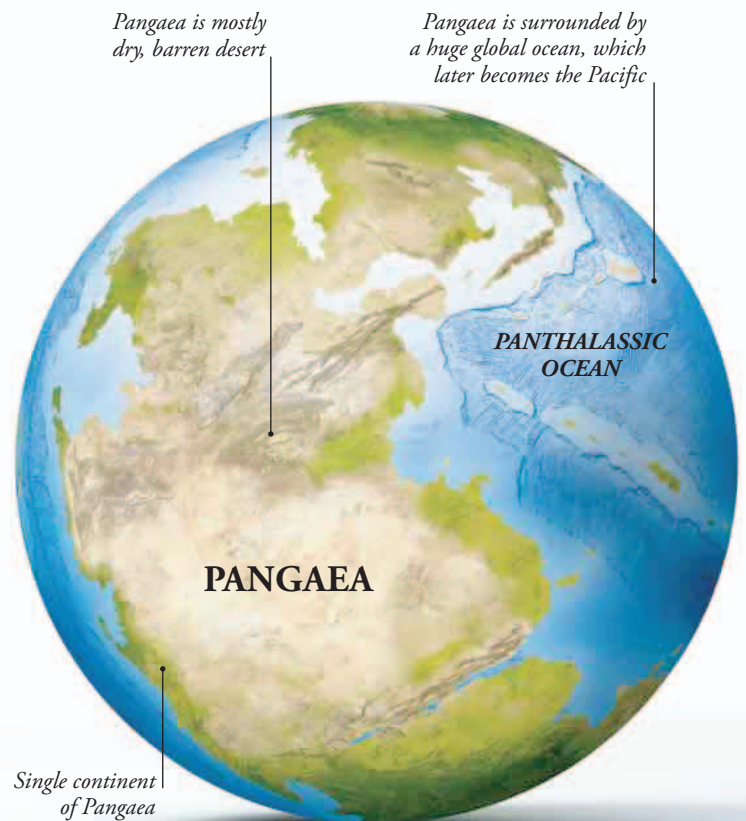


CONTINENTAL DRIFT

As oceans expand and contract, they push continents apart or draw them together. Over the 4.5 billion years of Earth's existence, this has changed the map of the world many times. Until about 100 million years ago, the continents would have been unrecognizable. It was only toward the end of the Mesozoic age of dinosaurs, about 66 million years ago, that the world as we know it began to form.

Evolving oceans

As fast as new ocean floor is created in some parts of the world, old ocean floor is destroyed in others. The two processes balance out, so the planet does not get any bigger or smaller. But this does mean that some oceans are expanding while others are shrinking. Over many millions of years, these movements shift the continents around the globe, pulling them apart to create new oceans, and pushing them together to squeeze older oceans out of existence.



▲ 250 MILLION YEARS AGO

At the beginning of the age of dinosaurs, 250 million years ago, all the land had been pushed together into a vast supercontinent.